

Airport(airport-id, iata, Nome, City, Country)

1NF

ALL value are atomic (No Multi-valued, No repeating group)

2NF

airport-id Primary key

Hence no ~~Functional~~ dependencies

3NF

Partial

Remove transitive dependency

airport-id $\rightarrow$ City

City $\rightarrow$ Country

Airport(airport-id, iata, Nome, City-id)

City (City-id, City-Nome, Country-id)

Country (Country-id, Country-Nome)

Airline (airline-id, iata-code, Name, country)

1NF

No multi-valued and no repeating groups

2NF

No ^{Partial} functional dependency (Primary key airline-id)

3NF

airline $\rightarrow$ country-id

country-id $\rightarrow$ country-name

airline (airline-id, iata-code, Name, country-id)

airline-country (country-id, country-name)

Aircraft (aircraft-id, model, total-seat, airline-id)

1NF

No multi-valued and no repeating groups

2NF

No ^{Partial} functional dependency (Primary-key aircraft-id)

3NF

aircraft-id $\rightarrow$ model-id

* model-id $\rightarrow$ model-name, total-seat

Aircraft (aircraft-id, model-id, airline-id)

Aircraft-model (model-id, model-name, total-seats)

=> Route(route-id, departure-airport, arrival-airport, distance)

1NF

No Multi-value or repeated group

2NF

No Partial dependency
because route-id is primary key
Routeid $\rightarrow$ departure-apt, arrival-apt, distance

3NF

No' Functional dependency

Routeid $\rightarrow$ departure-apt, arrival-apt, distance

Route(Routeid, departure-apt, arrival-apt, distance)

Seat(Seat-id, Seat-number, Seat-class, aircraft-id)

1NF

No Multi-value and No repeated groups

2NF

Seat-id $\rightarrow$ Seat-number, Seat-class, aircraft-id

No ~~functional~~ dependency because Seat-id is primary-key
partial

3NF

Seat-id $\rightarrow$ Seat-number, class-id, aircraft-id

Class-id $\rightarrow$ Seat-class

so

Seat(Seat-id, Seat-number, class-id, aircraft-id)

Seat-Class(class-id, Seat-~~Class~~ Name)

Passenger (passenger_id, name, passport_no, job)

1NF

Already in 1NF

2NF

passenger_id (Primary key)

Not partial dependency

3NF

No attribute depends on another
non-key attribute

No transitive dependency

Passenger (passenger_id, name, passport_no, job)

Flight(Flight-id, status, departure-time, arrival-time,
route-id, aircraft-id, airline-id)

1NF

No multi-value no repeated groups

2NF

Flight-id is primary key
Hence no partial dependency

Flight-id $\rightarrow$ status, departure-time, arrival-time, route-id
aircraft-id, airline-id

3NF

No Functional dependency
Hence

Flight(Flight-id, status, departure-time, arrival-time, route-id,
aircraft-id, airline-id)

Payment (Payment-id, amount, status, booking-id);

1NF

already in 1NF

No multivalued

No repeating Groups

2NF

Payment-id is Primary key

No Partial dependency

3NF

No transitive dependency

Payment (Payment-id, amount, status, booking-id)

Booking (booking-id, passenger-id, flight-id, seat-id)

1NF

already in 1NF

2NF

booking-id primary key

No partial dependency

3NF

No transitive dependency

No Non-key attribute determine any one
Non-key attribute

Hence

Booking (booking, passenger-id, flight-id,
seat-id);

Flight-Crew (Flight-id, Crew-id, role, Name)

1NF

already in 1NF

No multi-value

No repeating groups

2NF

Composite key (Flight-id, Crew-id)

Partial dependence

Crew-id $\rightarrow$ Name

(Flight-id, Crew-id) $\rightarrow$ role

3NF

No transitive dependency

Hence

Flight-Crew (Flight-id, Crew-id, role)

Crew (Crew-id, Name)

Baggage (baggage-id, weight, booking-id)

1NF

No Multi-valued
No repeating group

2nf

baggage-id $\rightarrow$ primary key
Hence No partial dependency

3NF

No transitive dependency

Hence

Baggage (baggage-id, weight, booking-id)

Airline DB

- 1 Country (country-id, country-name)
- 2 City (city-id, city-name, country-id)
- 3 Airport (airport-id, iata-code, name, city-id)
- 4 Airline (airline-id, iata-code, name, country-id)
- 5 Aircraft-model (model-id, model-name, total-seat)
- 6 Aircraft (aircraft-id, model-id, airline-id)
- 7 Seat-Class (class-id, class-name)
- 8 Seat (seat-id, seat-number, class-id, aircraft-id)
- 9 Route (route-id, departure-apt, arrival-apt, distance)
- 10 Flight (flight-id, status, departure-time, arrival-time, route-id, aircraft-id, airline-id)
- 11 Passenger (passenger-id, name, passport-no, dob)
- 12 Booking (booking-id, passenger-id, flight-id, seat-id)
- 13 Payment (payment-id, amount, status, booking-id)
- 14 Baggage (baggage-id, weight, booking-id)
- 15 Crew (crew-id, name)
- 16 Flight-Crew (Flight-id, crew-id, role)